



Assessment of the Accessibility of Potential Geosites and GIS-Based Design of Optimal Georoutes in Ardahan Province (Türkiye)

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Received: 18 October 2025 / Accepted: 5 January 2026 / Published online: 14 February 2026
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Abstract

Geotourism development in rural and mountainous regions requires an integrated understanding of how accessibility, scenic visibility, and digital awareness interact to shape visitor potential and spatial feasibility. This study presents a Geographic Information Systems (GIS)-based framework for assessing and optimizing georoutes in Ardahan Province, northeastern Türkiye—a region of outstanding geoheritage richness but limited infrastructural connectivity. Building upon the 2024 Ardahan Geopark Feasibility Report, seventy-three potential geosites were evaluated through spatial modelling that combined terrain impedance, network-based travel-time analysis, viewshed-based scenic visibility modelling, and digital representation metrics derived from Google Maps data. The normalized indicators were integrated into an Integrated Geoheritage Potential Index (IGPI) to quantify the relative significance of each geosite and support data-driven route design. Results reveal a pronounced spatial disparity: while most sites exhibit low to moderate accessibility and minimal digital visibility, a small cluster of geosites achieved high IGPI scores, forming a coherent “Çıldır–Kura Corridor” with strong geotourism potential. Route optimization, implemented through a network-based Traveling Salesman Problem (TSP) model, confirmed the feasibility of efficient one-day circuits linking high-priority sites, with total travel times ranging from 308 to 465 min. Additionally, a dedicated trekking route demonstrated the applicability of terrain-based modelling for adventure-oriented geotourism. The findings highlight that integrating spatial accessibility with scenic and digital visibility provides a powerful analytical tool for aligning geoheritage assessment with sustainable tourism and geopark planning. The proposed framework is replicable, transparent, and compatible with UNESCO Global Geopark principles, offering a transferable methodology for other rural regions seeking to transform dispersed geoheritage assets into connected and economically viable geotourism networks.

Keywords GIS-based modelling · Rural development · Accessibility analysis · Route optimization · UNESCO global geopark

Introduction

Tourism is a significant driver of regional development, yet it can also generate considerable ecological and social pressures if poorly managed (Jackson and Murphy 2006). This underscores the need for planning approaches that balance economic benefits with the conservation of natural

and cultural assets (Lee et al. 2013). Within this context, geotourism has emerged as a sustainable alternative that emphasizes abiotic nature while integrating ecological and cultural values through frameworks such as the ABC (Abiotic, Biotic, Cultural) model (Hose 1995; Dowling 2013; Ólafsdóttir and Dowling 2014; Ólafsdóttir and Tverijonaite 2018). The UNESCO Global Geopark programme (UGGp) further institutionalizes this philosophy by providing a governance framework that promotes geoheritage conservation, education, and community-oriented sustainable development (UNESCO 2016).

Effective geotourism planning increasingly relies on GIS-based tools, which have become indispensable for the inventorying, spatial analysis, and multi-criteria evaluation of geoheritage resources (Bahaire and Elliott-White 1999;

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Brilha 2016; Singh 2015; Acharya et al. 2022; Li et al. 2025). Within this framework, network analysis plays a central role in quantifying accessibility and optimizing visitor routes. Rooted in network theory—which provides critical insights into tourist flows and destination structures (Scott et al. 2008; Liu et al. 2017; Kang et al. 2018)—GIS-based network analysis enables the integration of travel-time modelling, service-area mapping, and optimization algorithms to identify efficient travel paths and enhance visitor experience. Such analyses directly support sustainable tourism infrastructure planning by reducing environmental impacts and improving spatial efficiency (Boers and Cottrell 2007). Techniques such as service-area and isochrone analysis allow for the measurement of spatial accessibility (Santana-Santana et al. 2020), while algorithms such as Least-Cost Path (LCP) and the TSP facilitate the design of itineraries that minimize travel cost or time while maximizing site coverage (Atkinson et al. 2005; Bagli et al. 2011; Effat et al. 2024). These methods have been successfully applied in diverse contexts, ranging from cross-boundary tourist flow analysis to geotrail optimization in post-mining landscapes (Williams et al. 2025). Although scenic value is a critical component of geotourism, establishing a spatial baseline of physical accessibility remains essential for operational planning and route design (Buhalis and Law 2008; Boley et al. 2011).

In Türkiye, geoheritage research has gained considerable momentum over the past decade, with studies conducted across a wide range of geological and geomorphological settings (Kazancı 2012; Cengiz et al. 2021; Ertekin et al. 2021; Mülayim et al. 2025; Güngör 2024; Akbulut et al. 2023; Erginal et al. 2021). Bibliometric analyses further confirm this expansion and diversification of research themes (Turoğlu 2025; Göktaş and Boyraz 2025). This growing body of work highlights Türkiye's substantial geopark potential, a conclusion reinforced by the recognition of Kula–Salihli as the country's first UNESCO Global Geopark (Demir and Aytaç 2018; Gümüş 2019). Alongside detailed inventories and qualitative assessments (Dölek and Şaroğlu 2017; Canpolat et al. 2020; Erginal et al. 2021), GIS-based route planning has emerged as a developing research focus (Karadeniz et al. 2022; Bilgiç et al. 2024; Gungor et al. 2025; Sucu and Mantaş 2025). However, most existing studies concentrate on relatively accessible or already-established destinations. For peripheral rural regions such as Ardahan, where infrastructural constraints and remoteness limit tourism visibility, accessibility analysis and route optimization are particularly critical. These approaches not only support geopark candidacy processes but also align with national rural development strategies by integrating geoheritage assets into sustainable tourism planning.

Ardahan Province was selected as the study area for several interrelated reasons. The region was recently the focus of the Ardahan Geopark Feasibility Project (Ardahan University 2024), supported by the Serhat Development Agency under the Feasibility Support Programme of the Republic of Türkiye Ministry of Industry and Technology. This project identified 73 potential geosites of geological and geomorphological significance, highlighting Ardahan's strong candidacy for UNESCO Global Geopark designation. Despite this valuable foundation, the feasibility report primarily addressed the distribution and typology of geoheritage resources, leaving unresolved the critical questions of how these sites can be spatially connected and which can be realistically incorporated into visitor itineraries given the province's mountainous terrain and dispersed settlement structure. Located within the Alpine–Himalayan orogenic belt, Ardahan hosts a wide range of geoheritage features, including glacial valleys, volcanic plateaus, tectonic depressions, and fluvial canyons. At the same time, its peripheral economic position, limited infrastructure, and ongoing out-migration trends underscore the need for data-driven strategies that integrate geoconservation with local economic revitalization.

To address this gap, the present study develops a GIS-based spatial modelling framework that bridges geoheritage inventorying and operational route planning. Building upon the Ardahan Geopark Feasibility Report (2024), the research integrates accessibility mapping, scenic visibility analysis, and network-based optimization modelling to evaluate the spatial feasibility of potential georoute scenarios. Guided by the principles of the UNESCO Global Geopark framework, the study aims to enhance regional accessibility, diversify local tourism offerings, and contribute to sustainable rural development. Specifically, it advances three analytical objectives:

- (1) evaluating the connectivity and travel-time feasibility of Ardahan's 73 geosites through network-based isochrone and service-area analyses;
- (2) assessing terrain-based and visual accessibility to determine the spatial potential of each geosite; and
- (3) optimizing georoute configurations using a TSP-based approach to design efficient and realistic visitor itineraries.

Methodologically, the research employs a multi-criteria GIS framework that integrates terrain impedance modelling, network-based travel-time analysis, scenic visibility, and digital awareness indicators. The resulting models provide a reproducible and data-driven approach to geosite evaluation and georoute planning. Ultimately, this study establishes a transferable methodological bridge between geoheritage

assessment and spatial planning, offering practical guidance for rural regions seeking UNESCO Global Geopark designation.

This study contributes to existing literature in three main respects. First, it introduces an IGPI that jointly evaluates physical accessibility, scenic visibility, and digital visibility—dimensions that have previously been examined largely in isolation within the Turkish context. Second, it incorporates viewshed-based scenic modelling and Google Maps-derived digital representation metrics, offering a novel visibility-oriented framework for regional geoheritage assessment. Third, by applying a network-based TSP model in a low-connectivity, high-relief environment, the study advances georoute optimization beyond heuristic or manually defined approaches. Together, these contributions provide a coherent methodological link between geoheritage inventorying and operational geotourism planning. It is important to note that the IGPI does not aim to reassess the intrinsic scientific or educational value of geosites, which has already been established in the official feasibility inventory. Instead, IGPI operationalizes spatial performance by integrating accessibility, scenic exposure, and digital representation under current infrastructural and perceptual conditions.

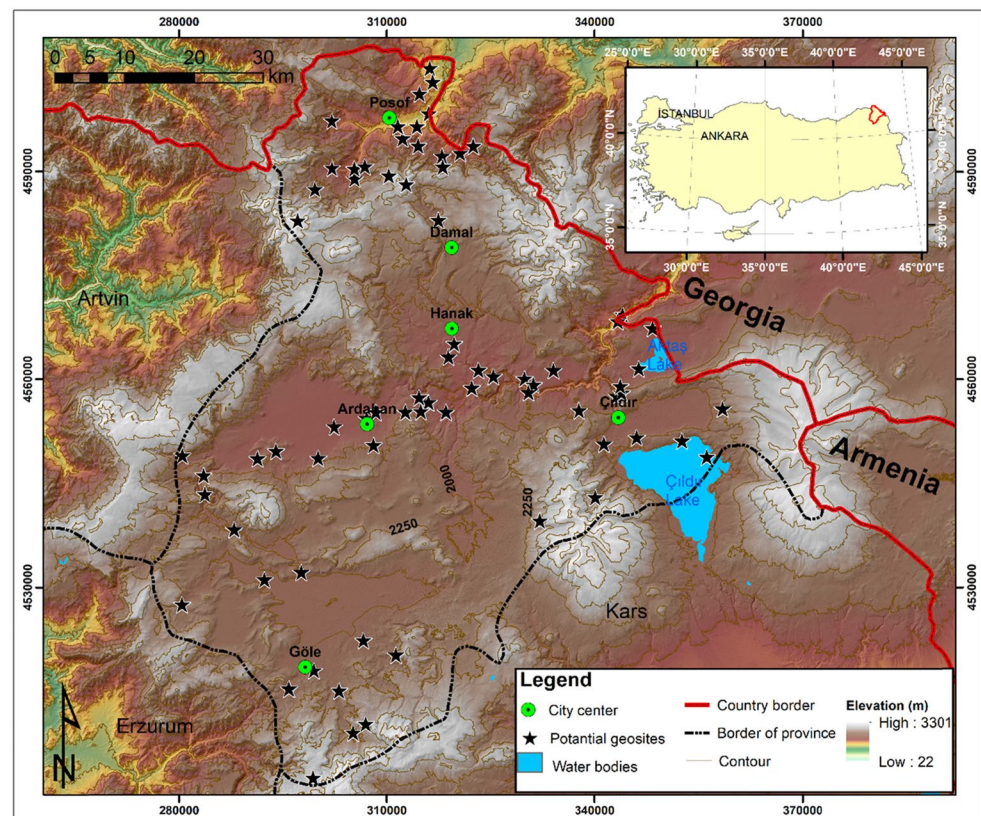
Data and Methods

Study Area

Ardahan is in northeastern Türkiye, covering an area of approximately 5,000 km². It borders Georgia and Armenia to the east, Artvin to the west, Erzurum to the southwest, and Kars to the south (Fig. 1). The Türkgözü and Aktaş border gates provide important commercial and transportation connections with Georgia (Ardahan University 2024). The province lies at an average elevation of 1,800 m, reaching over 3,000 m in the Allahuekber and Yalnızçam ranges and in volcanic massifs such as Cin, Kel, Ilgar, and Kısır mountains. Major depressions such as Göle, Ardahan, Çıldır, and Aktaş are of tectonic and alluvial origin and are linked by deeply incised valleys (Koçman 1981; Aygün and Akbulak 2017). The region experiences a harsh continental climate with long, cold winters and short, cool summers. Mean annual temperature (1958–2023) is 3.7 °C, with the lowest average in January (−11.2 °C) and the highest in July (16.0 °C). Mean annual precipitation is 555.8 mm, concentrated in spring and early summer.

(Turkish State Meteorological Service, 2025). The short growing season and frequent frosts limit agricultural

Fig. 1 Location of study area



productivity, while extensive pasturelands support livestock husbandry. Cereal production is dominated by barley and wheat, fodder crops are widespread, and limited fruit cultivation occurs in Posof (TURKSTAT, 2024a). Economically, the province depends heavily on agriculture and animal husbandry, which, combined with limited diversification and harsh climatic conditions, has led to long-term outmigration. Ardahan's 2023 population was 92,819, with a density of 19 persons/km², well below the national average (TURKSTAT, 2024b). This highlights the province's rural character and the close link between environmental constraints, economic structure, and demographic decline.

Data Sources

We assessed 73 potential geosites in Ardahan Province using integrated geospatial, environmental, and socio-cultural data. The geosite inventory was taken from the 2024 Ardahan Geopark Feasibility Report. All spatial layers were stored in a unified GIS geodatabase and projected to EPSG:32,638. Topography was derived from ALOS-PALSAR 30 m DEM

(JAXA 2015) to obtain elevation, slope, and terrain form. Land use/land cover followed CORINE 2018 (EEA). Administrative boundaries and hydrography were obtained from the General Directorate of Mapping (2021). The road and settlement layers were sourced from OpenStreetMap, a widely used open dataset for accessibility and network analyses (Zielstra and Hochmair 2011). Digital visibility was captured from Google Maps (user-generated content) as a proxy for online engagement, following recent tourism and geoheritage practice (Marine-Roig and Clavé 2015; Pereira et al. 2023; Fuchs et al. 2025). Data processing and harmonization were performed in ArcGIS Desktop 10.8 and Python 3.11 (GeoPandas, Rasterio) to ensure reproducibility and interoperability (Table 1).

Analytical Methods

The methodological framework was developed to quantify and spatially evaluate UGGp criteria through a combination of GIS-based spatial analyses and Python-driven modeling. The approach focuses on three measurable dimensions of geosite potential—accessibility, scenic visibility, and digital awareness—derived from geospatial and user-generated datasets. All analyses were applied to the 73 potential geosites identified in the Ardahan inventory. Each indicator was normalized and integrated into an IGPI, providing a comparative measure of overall site potential. Subsequently, network and route optimization analyses were conducted using OpenStreetMap (OSM) data processed via the OSMnx framework (Boeing 2017) and Google OR-Tools, to design feasible one-day georoutes linking high-potential sites under realistic travel-time constraints. This integrated framework ensures a quantitative, reproducible, and decision-oriented assessment, aligning spatial analysis outputs with UGGp planning principles and the sustainable development objectives of Ardahan's prospective geopark.

Accessibility Analysis

Accessibility was evaluated using an integrated spatial modelling framework combining terrain-based impedance and network-based travel-time modelling. This two-stage approach followed the methodological principles of Mikhailenko et al. (2021), Kubalíková (2013), and Reynard and Brilha (2017). In the first stage, local accessibility was estimated from terrain constraints. For each geosite, the minimum distance to the nearest OSM road segment was calculated, and the corresponding road class and average travel speed (10–90 km h⁻¹) were assigned according to OSM definitions. The topographic slope between the site and the road was derived from a 30 m DEM according to Eq. (1):

Table 1 Data sources used in the analysis

Dataset	Source	Date(s)	Spatial resolution/scale/data type
Geosite inventory (73 sites)	Feasibility report of Ardahan province	2024	Point, field survey, GIS database
Digital elevation model (terrain, slope, elevation)	ALOS-PALSAR DEM (Japan Aerospace Exploration Agency, https://search.asf.ala.ska.edu/)	2015 (latest release)	30 m, raster
Land cover	CORINE Land Cover (European Environment Agency, 2018 https://land.copernicus.eu/)	2018	100 m, raster
Roads	OpenStreetMap (https://www.openstreetmap.org/)	Accessed 2025	Vector
Settlement/village points	OpenStreetMap (place features), validated with (Republic of Türkiye Ministry of Interior, n.d.)	Accessed 2025	Point, vector
Administrative boundaries, hydrography	Republic of Turkey, Ministry of National Defence, General Directorate of Mapping (https://www.harita.gov.tr/)	2021	Vector
Population statistics	Turkish Statistical Institute (https://data.tuik.gov.tr/)	2024	Tabular (district level)
Social media data	Google Reviews	Accessed 2025	User-generated content (proxy indicators)

$$S = \frac{|Z_s - Z_r|}{D} \times 100 \quad (1)$$

where Z_s and Z_r are the site and road elevations (m) and D is the horizontal distance (m). Land-cover resistance was obtained from CORINE 2018 data, considering the dominant class within a 250 m buffer. Each land-cover type was assigned a friction coefficient reflecting relative movement difficulty, following Weiss et al. (2018). These parameters were combined into a terrain accessibility index using Eq. (2):

$$A_{\text{TERRAIN}} = e^{-(0.0004D + 0.01S + 0.05(LC - 1))} \quad (2)$$

where A_{TERRAIN} ranges from 0 (low) to 1 (high). Coefficients were adapted from Weiss et al. (2018) and calibrated for local conditions. Field observations confirmed the accuracy of accessibility classes (Vehicle–Easy, Vehicle–Short Walk, Walk–Moderate, Walk–Difficult, 4×4 Access, Remote Trekking). In the second stage, regional accessibility from Ardahan city center was calculated using a graph-based network model built from the OSM road dataset. The road network was first converted to a routable topological graph $G(V, E)$ using the open-source OSMnx framework (Boeing 2017), which standardizes the transformation of OpenStreetMap data into analyzable network structures compatible with NetworkX-based modelling. Travel time for each segment computed using Eq. (3):

$$T_{\text{edge}} = \frac{L}{V} \times 60 \quad (3)$$

where L is segment length (km) and V the mean speed (km h⁻¹). The shortest-time path between the city center and each geosite was determined using Dijkstra's algorithm, yielding driving time (T_{drive}) and distance (D_{drive}). For sites requiring walking access beyond the last drivable node, walking time (T_{walk}) was calculated using Tobler's hiking function (Eq. 4):

$$v_{\text{walk}} = 6e^{-3.5|\tan(\theta) + 0.05|} \quad (4)$$

and total access time was computed as $T_{\text{total}} = T_{\text{drive}} + T_{\text{walk}}$. Trailheads were defined where terrain impedance exceeded the 4×4-access threshold. Travel times were grouped into three isochronal accessibility zones—≤ 1 h (high), 1–2 h (moderate), and ≥ 2 h (low)—following Kolcsár and Szilassi (2018) and Gungor et al. (2024). Integrating the terrain-based (A_{TERRAIN}) and network-based (T_{total}) results enabled a multiscale assessment of both physical approachability and regional connectivity, providing a reproducible basis for georoute and infrastructure planning in the Ardahan.

Scenic Visibility Assessment

The scenic visibility of each potential geosite was quantified through a viewshed analysis implemented with WhiteboxTools and complementary Python libraries. This method determines how extensively each site is visually exposed within its surrounding terrain, providing a reproducible indicator of scenic prominence and perceptual accessibility (Stucky 1998; Elkaichi et al. 2023; Martínez-Murillo et al. 2025). For each geosite, a circular buffer with a 5 km radius was defined to represent its local visual catchment area, consistent with previous visibility studies where 3–5 km extents effectively capture near- and mid-range topographic prominence (Gillings and Wheatley 2020). The 30 m DEM of Ardahan Province was clipped to each buffer extent to reduce computational load while maintaining local topographic accuracy. Using the viewshed function of WhiteboxTools, visibility was computed from the geosite coordinate as the observer point with an eye height of 1.7 m, corresponding to an average standing observer. Each output raster classified visible and non-visible cells (1 = visible, 0 = non-visible). Raster data were post-processed in Python using *Rasterio* and *NumPy* to calculate the total number of visible cells (N_{visible}) and total analysis cells (N_{total}). The scenic visibility index (V_{SCORE}) was derived according to Eq. (5):

$$V_{\text{SCORE}} = \frac{N_{\text{visible}}}{N_{\text{total}}} \times 100 \quad (5)$$

where V_{SCORE} expresses the proportion of terrain visible from the site, in %. Higher values indicate more open or elevated landscapes, whereas lower values represent enclosed or shielded settings. All calculations were performed under uniform DEM conditions (excluding vegetation or built-structure masking) to ensure comparability among sites. The resulting V_{SCORE} values provided a standardized measure of scenic visibility, which was later integrated with accessibility and digital visibility indicators in the composite geosite assessment framework. The viewshed-based visibility index represents topographic openness derived from a bare-earth DEM and does not account for vegetation cover or built structures. Therefore, it should be interpreted as a relative indicator of potential scenic exposure rather than an exact measure of on-site visual experience.

Digital Visibility Analysis

The digital visibility of each geosite was assessed using publicly available Google Maps data as a proxy for online recognition and engagement with geoheritage and tourism sites (Girardin et al. 2008; Munawir et al. 2019; Leiras and

Eusébio 2024; Fuchs et al. 2025). Digital visibility was not interpreted as a direct measure of visitation intensity, but as an indicator of representational presence within dominant digital navigation platforms that shape pre-visit awareness and route selection. This approach is consistent with studies highlighting the analytical usefulness of user-generated geographic content for assessing destination awareness and perceived accessibility. For each of the 73 geosites, two universally available indicators were collected: the number of user reviews (R) and the average star rating (S) on a 1–5 scale. Other Google Maps metrics (e.g., “Popular Times” or mobility-based activity indicators) were excluded due to their inconsistent availability for rural and low-traffic locations, ensuring methodological comparability across sites. Both indicators were normalized to a 0–1 range using min–max transformation, with geosites lacking Google Maps records assigned a value of zero. A composite Digital Visibility Index (DVI) was then calculated according to Eq. (6):

$$DVI = 0.6 \frac{R_n}{R_{max}} + 0.4 \frac{S_n}{S_{max}} \quad (6)$$

where R_n and S_n denote the normalized review count and rating, respectively. The weighting scheme reflects the stronger influence of interaction frequency relative to average user perception, following established practices in digital geography studies (Putri et al. 2025; Al Beruni et al. 2025). Higher DVI values indicate geosites with greater online recognition and engagement, while lower values reflect limited digital representation. Geosites assigned a value of zero should therefore be interpreted as digitally underrepresented within Google Maps, reflecting platform bias and rural content asymmetries rather than a lack of geoheritage significance.

Defining Core Geosites for Georoute Design

Core geosites forming the backbone of the proposed georoute were identified by integrating three quantitative indicators: accessibility (A), scenic visibility (V), and digital visibility (Pd). Together, these indicators capture the functional, aesthetic, and representational dimensions shaping geosite connectivity and visitor experience (Reynard and Brilha 2017; Mikhailenko et al. 2021). Accessibility was treated as a threshold condition, as sites that cannot be safely reached are effectively excluded from most visitor itineraries regardless of their intrinsic value (Brilha 2016). Scenic visibility and digital representation play complementary roles by enhancing in situ interpretation and shaping pre-visit awareness and route choice (Dowling and Newsome 2017; Martínez-Graña et al. 2017). The 73 candidate geosites were adopted from the official Ardahan Geopark feasibility

inventory, in which their scientific and geocultural values had already been assessed. Accordingly, the present analysis focused on spatial performance and interconnectivity rather than intrinsic geoheritage significance. All indicators were normalized to a 0–1 range using min–max scaling, with accessibility values inverted so that higher scores indicate greater ease of access. The normalized indicators were then aggregated using an unweighted arithmetic mean to derive the Integrated Geoheritage Potential Index (IGPI) (Eq. 7):

$$IGPI = \frac{A + V + Pd}{3} \quad (7)$$

Equal weighting was intentionally applied to ensure transparency and to minimize subjective bias in indicator influence, in line with established composite-index design principles emphasizing neutrality and reproducibility (Booyesen 2002; OECD 2008; Greco et al. 2019). Although accessibility is often regarded as the dominant factor in geotourism feasibility, assigning a higher weight would require empirically unvalidated assumptions for the Ardahan context. Therefore, accessibility, scenic visibility, and digital visibility were treated as complementary dimensions and weighed equally. The resulting Integrated IGPI provides a balanced measure of geosite performance by jointly capturing spatial and perceptual accessibility. Higher IGPI values indicate geosites that are well connected, visually prominent, and digitally recognized, thus representing optimal core nodes for georoute planning. The robustness of the equal-weighting scheme was evaluated through Spearman’s rank correlation analysis across alternative weighting scenarios, a standard sensitivity-testing approach in composite index studies (Zar 1999), which demonstrated high stability in geosite rankings (Table 4). Consequently, equal weighting does not bias or distort IGPI outcomes and supports transparent, reproducible route design consistent with international geoheritage assessment practices.

Route Optimization

Route optimization aimed to identify the most efficient travel sequence among the selected core geosites while minimizing total travel time. OSM road network was converted into a topological graph structure using the NetworkX library, where only driveable road classes (motorway, primary, secondary, tertiary, residential) were retained to ensure realistic routing. Each edge was assigned a travel impedance value derived from segment length and its corresponding mean speed, according to Eq. (8):

$$T_{edge} = \frac{L}{V} \times 60 \quad (8)$$

where L is the road segment length (km) and V is the mean speed (km h^{-1}). Speeds were assigned by OSM rural road classes. Shortest-time paths from Ardahan city center to each geosite were computed with Dijkstra on the OSMnx graph, yielding a symmetric time–distance matrix. This matrix fed a TSP for a closed tour, solved with a 2-opt heuristic to obtain a near-optimal route (Reinelt 2001; Applegate et al. 2011). Cumulative travel time combined driving, short approach walks, and standardized waiting (30 min/site). The use of a standardized waiting time is consistent with common practice in TSP-based tourist trip design models, where uniform visit durations are employed to ensure comparability across itineraries and to avoid unnecessary behavioural assumptions (Gavalas et al. 2014; Porras et al. 2022). For trekking scenarios, vehicular impedance was replaced by terrain cost surfaces (slope+land cover); walking speed followed Tobler’s hiking function to model on-foot times between trailheads and targets. The optimized outputs support implementable day- and multi-day georoutes, balancing accessibility and efficiency, and the workflow is readily transferable to other candidate geopark regions.

Results

All site-level metrics are compiled in Table S1 (Supplementary Information), the main text reports only priority results.

Accessibility Analysis

Two complementary analyses were conducted to assess the accessibility of potential geosites across Ardahan Province. The first focused on physical accessibility classes, integrating three key spatial parameters—road proximity, slope along the access path, and land-cover-based terrain resistance derived from the CORINE 2018 dataset. Accessibility analysis revealed pronounced spatial and functional variability among Ardahan’s 73 potential geosites. As illustrated in Fig. 2, nearly one-third of the sites (31.5%) are classified as Vehicle–Easy access, indicating direct road connection and minimal topographic constraint. A further 26.0% fall within the Vehicle–Short walk class, where access involves

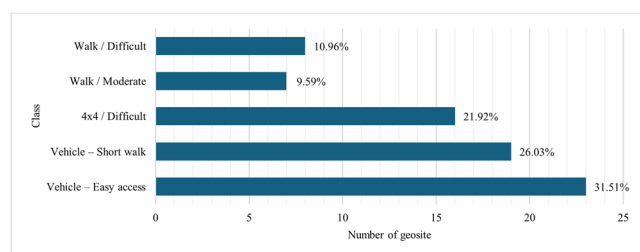


Fig. 2 Distribution of potential geosites by accessibility class (%)

a brief on-foot approach (<500 m). In contrast, 4×4/Difficult sites represent 21.9% of the total, typically located along unpaved mountain or forest roads that restrict ordinary vehicle use. The remaining classes—Walk/Moderate (9.6%) and Walk/Difficult (11.0%)—reflect sites requiring hiking over rugged terrain, highlighting accessibility limitations in upland and remote valleys (Fig. 2).

The spatial distribution of these classes underscores the influence of both topography and transportation infrastructure. The most accessible geosites are concentrated along the Ardahan–Çıldır corridor and within low-slope basins near Göle and Hanak, where primary and secondary roads provide direct connectivity. Conversely, difficult-access sites cluster in the Posof Mountains and the high-elevation ridges surrounding Aktaş and Çıldır Lakes. These peripheral areas coincide with steep slopes and limited road density, suggesting that geomorphological constraints and sparse settlement networks significantly reduce mobility potential (Fig. 3).

Travel-time analysis from Ardahan city center (Fig. 4) further quantifies this spatial disparity. More than two-thirds of all sites ($n=53$) are reachable within ≤ 1 h, reflecting a relatively high level of regional accessibility for day-trip visitors. Sixteen sites (22%) fall within the 1–2 h isochrone, while only four sites (6%) require travel times exceeding 2 h, all of which are located in mountainous sectors near the international border. This pattern is clearly reflected in the accessibility zones shown in Fig. 4, demonstrating that despite Ardahan’s rugged relief, the existing road network allows efficient access to most high-potential geosites, particularly those situated near main valleys and transportation corridors.

Overall, the accessibility pattern reveals a clear duality: a central–eastern belt of easily reachable sites that can form the backbone of visitor routes, and a northern–southern highland fringe where access remains challenging but offers opportunities for specialized eco- and trekking-based geo-tourism. This differentiation provides a spatial foundation for the subsequent georoute optimization phase, ensuring that route design integrates both logistical feasibility and the representativeness of Ardahan’s diverse geoheritage.

Scenic Visibility Results

The scenic visibility results revealed substantial spatial heterogeneity in the scenic visibility of the 73 potential geosites across Ardahan Province. The calculated visible-area values ranged between 0.15 km^2 and 28.32 km^2 , while normalized scenic visibility (V_{SCORE}) varied from 0.2 to 36.1. This broad range reflects pronounced differences in topography, elevation, and surrounding landform openness. As illustrated in Fig. 5, most sites ($n=50$) fall below

Fig. 3 Spatial distribution of accessibility classes across Ardahan Province

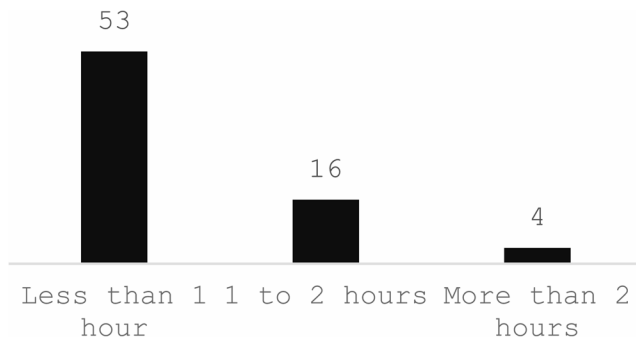
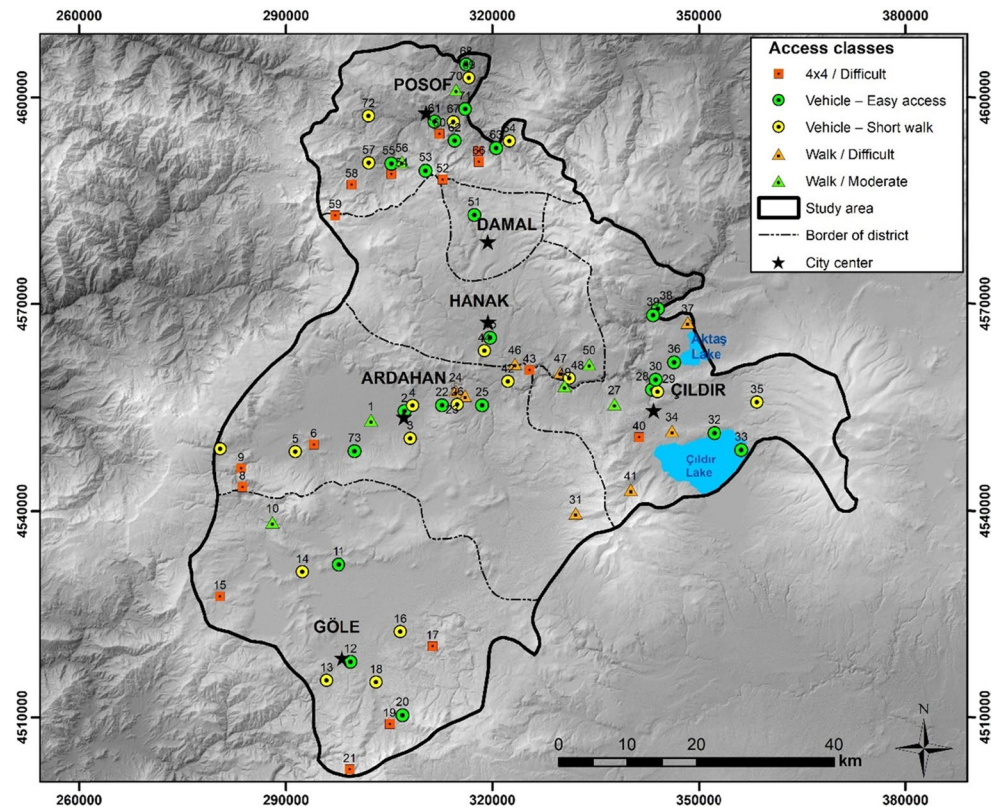


Fig. 4 Distribution of geosites by accessibility zones in Ardahan Province

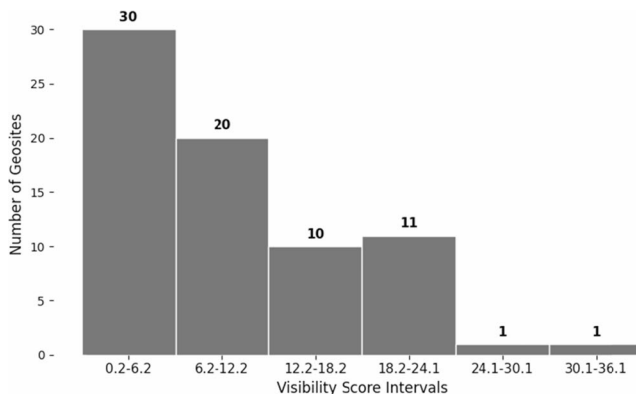


Fig. 5 Distribution of potential geosites by visibility-score intervals

a scenic visibility of 12.2, indicating that low to moderate visual exposure predominates throughout the province.

The highest visibility values occurred in open plateau and lacustrine environments, where low relief and unobstructed horizons allow extensive landscape perception. Sites such as Akçakale Island, Aktaş Lake and Tectonic Depression, Aşıkzülali Dome, and Binbaşak Fault Zone exhibited the strongest scenic dominance, each positioned on broad valley basins or elevated structural domes. These landforms combine visual prominence with geological distinctiveness, enhancing their attractiveness for interpretation and visitor experience. In contrast, low-visibility sites are situated within enclosed topographic settings—glacial cirques, narrow canyons, and deeply incised valleys—where steep slopes and vegetation substantially reduce the visible horizon. Typical examples include Kalaçatepe Crater, Göze Mountain Glacial Valley, and Harosman Caves. Although these localities possess considerable scientific and educational importance, their restricted visual openness limits their suitability for panoramic viewing or landscape interpretation purposes. Spatially, high-visibility geosites are concentrated in the central-eastern belt of Ardahan, particularly around Çıldır, Aktaş, and Aşıkzülali, where structural plateaus transition into wide tectonic depressions. In contrast, the western and northern highlands, notably within the Göze–Yalnızçam Massif and the Posof Mountains, exhibit

markedly reduced visibility due to pronounced relief and shadowed valley systems.

Overall, the scenic-visibility assessment indicates that approximately one-third of Ardahan's potential geosites display strong to very strong visual prominence. These visually dominant locations represent key landscape anchors with high interpretive and experiential value for geotourism development. Integrating them into future georoute designs will enhance both the aesthetic diversity and the visitor appeal of the regional geoheritage network, while less visible sites can continue to reinforce the scientific depth and representativeness of the overall geosite inventory.

Digital Visibility Analysis

The digital visibility assessment, based on Google Maps ratings, review counts, and activity levels, reveals considerable variation among the 73 geosites in Ardahan Province. The calculated DVI values range from 0 to 0.97, indicating a wide spectrum of online representation and public engagement. Approximately 70% of the geosites are either invisible or inactive, showing no user-generated content, reviews, or ratings (Table 2). These locations are primarily rural or scientifically significant sites with limited tourism exposure, such as geological formations and tectonic features that remain absent from digital tourism platforms. Their lack of digital representation suggests minimal awareness among both local communities and visitors.

Table 2 Categorization of geosites based on their digital visibility and user interaction levels on Google maps

Category	Definition	Number of Geosites (n)	Percentage (%)	SITE_NO	Representative Examples
Not visible on Google Maps	Geosites not listed or not visible on Google Maps; no digital record or user interaction (Visibility=None, SVI=0).	45	61.64	1, 3, 5, 6, 8, 11, 12, 13, 16, 17, 18, 19, 20, 21, 25, 26, 29, 30, 31, 33, 34, 35, 37, 38, 39, 40, 43, 45, 47, 48, 50, 55, 56, 57, 59, 62, 65, 66, 67, 68, 69, 70, 71, 72, 73	Göle Historical Dam Structure and Lake Barrier, Sarne Junction Valley and Hot Spring, Akçakale Island and Ancient Settlement Remains, Yolağzı Landslide Lakes
Listed but No Interaction	Geosites listed on Google Maps but without any user reviews or ratings (Visibility=Passive).	5	6.85	14,22,54,63,64	Dedeşen Village and Ancient Settlement Remains, Putka Lake, Posof Land Use Area, Savaşır Village Active Landslide Site, Süngülü Village Paleo-Landslide Site
Listed and Reviewed (Active)	Geosites with active Google Maps records including reviews, ratings, and user engagement (Visibility=Active).	23	31.51	2, 4, 7, 9, 10, 15, 23, 24, 27, 28, 32, 36, 41, 42, 44, 46, 49, 51, 52, 53, 58, 60, 61	Ardahan Castle, Kura Canyon External Caves, Bağdeşen Castle, Bülbülan/Kura Formation, Pseudokarstic Caves, Fairy Chimney Formations, Kalecik Castle and Basalt Columns, Canibeg Plateau, Kura Canyon and Incised Meanders

Overall, the findings highlight a pronounced digital visibility gap between accessible, tourist-oriented sites and remote or scientifically valuable but digitally underrepresented ones. Enhancing the online visibility of the latter group through targeted geotourism promotion and digital heritage mapping would be essential for a balanced geopark strategy in Ardahan.

The distribution of active geosites shows a clear spatial concentration along the Ardahan–Çıldır–Posof corridor, where accessibility and visual appeal overlap. A total of 23 geosites are actively listed and reviewed on Google Maps, reflecting notable online engagement. High scenic visibility values occur mainly around Çıldır Lake, Göze Mountain, Kolşik Forests, and Kalaçatepe Volcano, which stand out as visually dominant landscapes. Moderate visibility is observed at Ulgar Cone, Soğukpınar, Damal Atatürk Silhouette, and Kısırdağ Valleys, while sites in the central Ardahan basin such as Kalecik Castle and Bülbülan Formation show lower visibility. Overall, the results indicate that geosites with higher visual prominence and easier access tend to have stronger digital presence and interaction, emphasizing the link between landscape visibility and public engagement (Fig. 6).

Integrated Analysis

The application of the equally weighted integrated importance index (IGPI) reveals a clear hierarchy among the

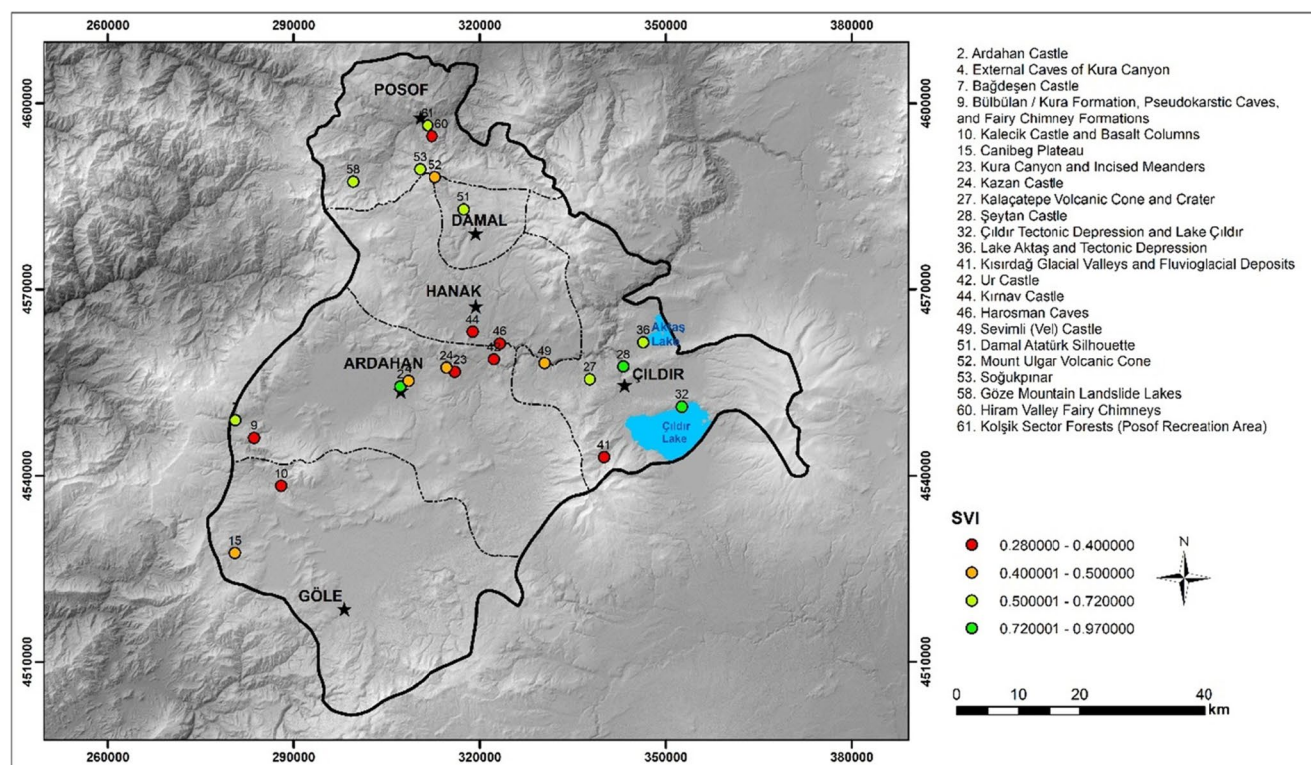


Fig. 6 Spatial distribution of geosites with active Google Maps records in the Ardahan

Ardahan geosites, with the majority ($n=67$) clustering within the Low and Moderate importance classes (Table 3). This distribution underscores a critical finding: despite their intrinsic geoscientific value, a significant number of geosites suffer from underrepresentation, primarily driven by constrained physical accessibility and negligible public digital visibility. The spatial isolation of these sites, often situated in remote or mountainous terrain with poor road connectivity, is a direct and major contributing factor to their depressed index values. In stark contrast, six geosites emerge as regional priorities, classified within the High and Very High importance tiers. These key sites—Ardahan Castle, Çıldır Tectonic Depression and Lake, Aktaş Lake and Tectonic Depression,

Akçakale Island, Kolşik Sector Forests (Posof Recreation Area), and Şeytan Castle—distinguish themselves through a robust combination of high accessibility, strong scenic value, and significant digital presence. This synergy results in the highest integrated scores in the inventory.

Notably, the spatial distribution of these top-tier geosites is non-random, forming a distinct and strategic “Çıldır-Kura Development Corridor” that arcs from the Çıldır Lake Basin southward. This linear alignment highlights a convergence zone of high-value natural and cultural heritage assets. Consequently, this corridor is identified as the primary candidate for focused geotourism development, landscape interpretation, and infrastructural investment, offering the greatest potential

Table 3 Statistical summary of geosite importance classes (Equal weighting Method)

Importance Class	Number of Geosites	Mean A_SCORE*	Mean V_SCORE**	Mean PD_SCORE***	Mean IGPI****	Representative Accessibility	Remarks
Very High	1	0.88	0.54	0.97	0.79	Urban and easily accessible	High visibility, strong cultural significance
High	5	0.87	0.47	0.67	0.69	Located along Çıldır–Aktaş corridor	Combination of good access and scenic diversity
Moderate	15	0.78	0.36	0.32	0.47	Partially accessible via secondary roads	Moderate accessibility, limited digital visibility
Low	52	0.63	0.27	0.12	0.29	Remote or mountainous	Poor accessibility and minimal visibility

* accessibility index; ** scenic value; *** public digital visibility; **** equally weighted integrated importance index

Table 4 Sensitivity analysis of the IGPI under alternative weighting schemes

Scenario	Weighting (A, V, Pd)	Spearman's ρ	p -value	Interpretation
S ₁ – Visual focus	(0.4, 0.4, 0.2)	0.957	<0.001	Very high stability
S ₂ – Access focus	(0.6, 0.25, 0.15)	0.869	<0.001	Moderate stability
S ₃ – Digital focus	(0.45, 0.3, 0.25)	0.975	<0.001	Extremely stable
S ₄ – Balanced	(0.5, 0.35, 0.15)	0.906	<0.001	High stability

for immediate and sustainable returns due to its inherent advantages in accessibility, scenic diversity, and existing recognition.

To assess the robustness of the IGPI and ensure that the observed hierarchy of geosites was not influenced by weighting bias, a sensitivity analysis was conducted. Four alternative weighting configurations emphasizing accessibility, visual exposure, and digital representation were tested. Spearman's rank correlation coefficients between the baseline (equal weighting) and these alternative scenarios ranged from 0.87 to 0.98 ($p < 0.001$), confirming that the overall ranking of geosites remained highly stable and insensitive to moderate variations in weighting assumptions. This finding validates the methodological reliability of the IGPI framework and supports the consistency of the derived importance hierarchy (Table 4).

Optimized Geotourism Routes

The route optimization analysis identified two complementary geotourism circuits designed to maximize spatial connectivity and travel efficiency among the potential geosite points of Ardahan Province (Fig. 7). The optimized configuration defines an integrated access network linking potential geosites through the existing primary and secondary road infrastructure, thereby ensuring continuity between the province's main tectonic, lacustrine, and volcanic geosystems. The model incorporates both vehicular and trekking components, providing an applicable framework for day trips and short-term geotourism itineraries.

The first configuration, GeoRoute-1 (Fig. 7), was developed as a full-day vehicular circuit connecting six potential geosite points with high and very high integrated potential values. As illustrated in Fig. 7, the route starts from Ardahan City Center and proceeds sequentially through Ardahan Castle (Fig. 8), Şeytan Castle (Fig. 9), Çıldır Tectonic Depression and Lake Çıldır (Fig. 10), Akçakale Island and Ancient Settlement Remains (Fig. 12), Aktaş Lake and Tectonic Depression (Fig. 11), and the Colchic Sector Forests located within the Posof Recreation Area (Fig. 13). Including standardized 30-minute stops at each site, the total cumulative travel time was calculated as 465.1 min (Table 5a). This route represents a full-day itinerary encompassing

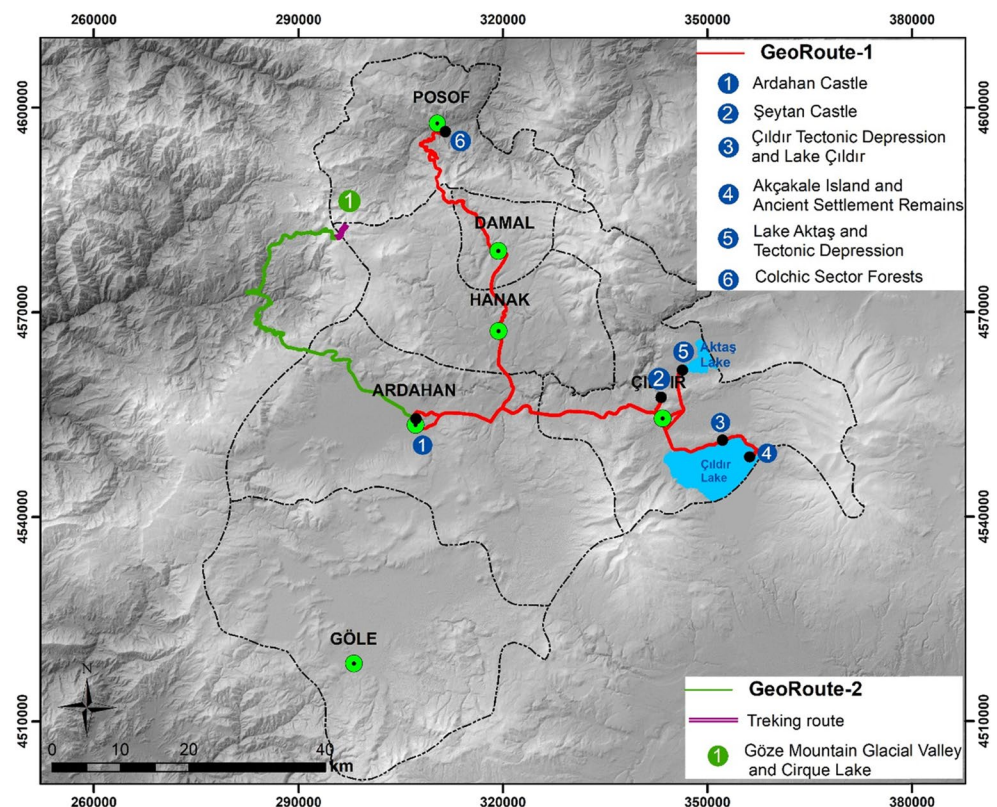
Fig. 7 Suggested georoutes and access network in the Ardahan




Fig. 8 Ardahan Castle



Fig. 9 Şeytan Castle



Fig. 10 View of Lake Çıldır and Akçakale Island



Fig. 11 Aktaş Lake



Fig. 12 View of Lake Çıldır from Akçakale Island



Fig. 13 Colchic Sector Forests (Posof Recreation Area) – Photo: Ardahan Geopark Feasibility Report

Ardahan's east–west geomorphological continuum and is suitable for vehicle-based visitor excursions.

Due to the peripheral location of Posof and the extended travel time required to reach the Colchic Sector Forests, an alternative configuration was produced by excluding

Table 5 Comparative travel-time sequences of the full and reduced optimum georoutes in the Ardahan

(a) Sequence	From	To	Drive_min	Walk_min	Wait_min	Access_tot	Cumu- lative
1	City center	Ardahan Castle	1.3	0	30	31.3	31.3
2	Ardahan Castle	Şeytan Castle	37.9	23	30	90.9	122.2
3	Şeytan Castle	Çıldır Tektonik Depresyonu ve Çıldır Gölü	41.6	0	30	71.6	193.8
4	Çıldır Tectonic Depression and Lake Çıldır	Akçakale Island and Ancient Settlement Remains	8.3	0	30	38.3	232.1
5	Akçakale Island and Ancient Settlement Remains	Aktaş Lake and Tectonic Depression	27	0	30	57	289.1
6	Aktaş Lake and Tectonic Depression	Colchic Sector Forests (Posof Recreation Area)	81.9	0	30	111.9	401
7	Colchic Sector Forests (Posof Recreation Area)	City center	64.1	0	0	64.1	465.1
(b)							
1	City center	Ardahan Castle	1.3	0	30	31.3	31.3
2	Ardahan Castle	Şeytan Castle	37.9	23	30	90.9	122.2
3	Şeytan Castle	Aktaş Lake and Tectonic Depression	17.2	0	30	47.2	169.4
4	Aktaş Lake and Tectonic Depression	Akçakale Island and Ancient Settlement Remains	27	0	30	57	226.4
5	Akçakale Island and Ancient Settlement Remains	Çıldır Tectonic Depression and Lake Çıldır	8.3	0	30	38.3	264.7
6	Çıldır Tectonic Depression and Lake Çıldır	City center	43.5	0	0	43.5	308.2

(a) Full route including the Colchic Sector Forests (Posof Recreation Area); (b) Reduced route excluding Posof. Travel times represent the sum of driving, walking, and on-site waiting durations (minutes). Cumulative totals correspond to total elapsed time from the Ardahan City Center

this western segment. As reflected in Fig. 7 and quantified in Table 5b, the reduced-core circuit connects five major potential geosite points—Ardahan Castle, Şeytan Castle, Aktaş Lake, Akçakale Island, and the Çıldır Tectonic Depression and Lake Çıldır—with a total cumulative travel time of 308.2 min. This configuration preserves the thematic and geomorphological representativeness of the full route while offering a more time-efficient alternative focused on the Ardahan–Çıldır corridor, making it a realistic option for single-day tours.

In addition to the vehicular circuits, a separate trekking route was delineated toward the Göze Mountain Glacial Valley and Cirque Lake, located in the southwestern part of the province. The itinerary involves a 77.6 km vehicle transfer from Ardahan City (140 min) to the mountain trailhead, followed by a 3.3 km uphill hike (55 min) to the cirque basin and a return along the same path. Including the downhill segment (50 min) and the return drive (140 min), the total cumulative travel duration equals 385 min. The proposed trekking route is designed as an out-and-back path due to topographic limitations and conservation considerations, ensuring both safety and minimal environmental impact. This route provides a spatially coherent one-day field excursion offering access to high-altitude glacial landforms and complements the vehicular georoutes within Ardahan's overall geotourism framework (Fig. 7; Table 6; Fig. 14).

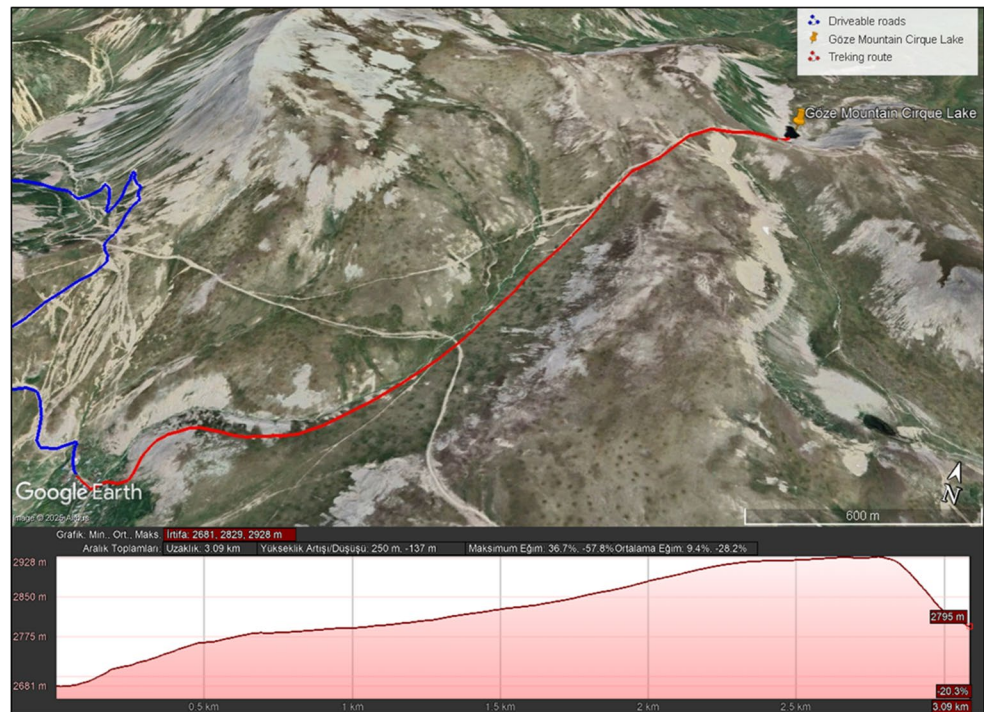
Discussion

Spatial Implications of Accessibility and Visibility

The results demonstrate that accessibility, scenic visibility, and digital representation jointly shape the spatial feasibility and geotourism potential of the seventy-three inventoried geosites in Ardahan Province. By integrating terrain impedance, network-based travel time, and digital visibility indicators, the study moves beyond single-parameter suitability assessments and reveals how spatial inequalities structure geoheritage accessibility in peripheral regions. While a limited number of geosites—mainly concentrated along the Ardahan–Çıldır axis—exhibit high physical accessibility and public visibility, the majority remain physically isolated and digitally underrepresented, constraining their integration into regional tourism systems. This contrast reflects a structural divide between core and peripheral landscapes, a recurrent challenge in rural geopark planning where scientific value does not automatically translate into public engagement or economic utilization (Farsani et al. 2014; UNESCO 2016). From a spatial perspective, accessibility patterns are primarily governed by topography and road hierarchy. Geosites located within low-slope basins or along primary and secondary roads generally require less than one hour of travel time, whereas those situated

Table 6 Access route characteristics for the Göze mountain glacial Valley and cirque lake

Sequence	From	To	Mode of travel	Distance (km)	Travel time (min)	Cumulative	Remarks
1	Ardahan City Center	Trailhead near Göze Mountain	Vehicle (driveable roads)	77.57	140	140	Mountain road drive via secondary and local roads; partially unpaved, limited access beyond 2000 m elevation.
2	Trailhead (Göze Mountain)	Cirque Lake	trekking	3.3	55	195	Moderate–steep hiking trail across glacial valley slopes; uphill segment to cirque basin.
3	Cirque Lake	Trailhead (return)	trekking	3.3	50	245	Easier downhill section; same route used for return.
4	Trailhead (Göze Mountain)	Ardahan City Center	Vehicle (return)	77.57	140	385	Same mountain route for return travel.

Fig. 14 Trekking route to the Göze Mountain Cirque Lake and corresponding elevation profile

in mountainous peripheral sectors such as Posof and Göze demand travel times exceeding three hours. This east–west gradient, extending from relatively connected basin landscapes to remote uplands, aligns with network theory principles emphasizing node centrality and corridor connectivity as key determinants of rural destination hierarchies (Lee et al. 2013). Similar corridor-dependent accessibility structures have been documented in Türkiye, where geosite reachability closely follows road density and topographic constraints, reinforcing the role of transport corridors as the spatial backbone of geotourism development (Karadeniz et al. 2022; Gungor et al. 2025). Internationally, comparable spatial logics have been observed in peripheral geotourism contexts. For instance, Herrera-Franco et al. (2022) show that high-value geosites along Ecuador’s Pacific Coast cluster along the Guayas–Santa Elena transport axis, forming linear georoutes where accessibility and thematic coherence

reinforce each other. These parallels suggest that the emergence of the Ardahan–Çıldır corridor reflects a broader spatial regularity: geotourism systems tend to consolidate along accessible linear pathways, while remote mountainous landscapes remain structurally disadvantaged despite their geomorphological richness.

Integration of Physical and Digital Visibility

Scenic visibility analysis complements the accessibility findings by highlighting marked contrasts in perceptual exposure across the study area. Open lacustrine and volcanic landscapes, such as Çıldır and Aktaş Lakes, display high scenic dominance and long-distance visibility, whereas enclosed glacial cirques, narrow valleys, and canyon systems exhibit limited visual exposure. These differences underscore the importance of visibility mapping as an interpretive tool for

route design and visitor engagement, particularly in landscapes where geomorphological significance is not immediately legible to non-expert audiences (Martínez-Graña et al. 2017). When high scenic visibility coincides with strong accessibility, geosites tend to generate substantially higher levels of online engagement. In Ardahan, however, approximately 62% of inventoried geosites currently lack any representation on Google Maps (data accessed May 2025), revealing a pronounced digital divide. Rather than serving as a proxy for actual visitation, digital visibility in this context reflects representational capacity—namely, the extent to which geosites are embedded in digital knowledge infrastructures that mediate awareness, navigation, and itinerary formation. Similar patterns have been reported in rural geopark contexts, where digital tools have been shown to partially offset physical remoteness by enhancing perceptual and informational accessibility (Martínez-Graña et al. 2017; Deng and Zou 2022). These findings indicate that in peripheral regions such as Ardahan, digital visibility functions as a structural component of geotourism systems rather than a supplementary communication layer. Strengthening digital representation—through open geosite portals, participatory mapping, and locally generated content—thus emerges as a necessary condition for integrating visually constrained or physically remote geosites into coherent georoute narratives (Gavilanes Montoya et al. 2021).

Multi-Criteria Integration and Route Optimization

IGPI constructed using equally weighted accessibility, scenic visibility, and digital visibility components, provides a balanced and reproducible framework for evaluating geosite performance. The results identify six geosites—Ardahan Castle, Şeytan Castle, Aktaş Lake, Çıldır Lake, Akçakale Island, and the Colchic Sector Forests—as achieving high or very high performance scores, collectively forming a coherent Çıldır–Kura axis. This cluster links tectonic depressions, lacustrine plains, and volcanic landscapes within a spatially continuous corridor, reinforcing the role of accessibility corridors as organizing structures for geotourism systems. Unlike conventional AHP-based suitability models, which rely on expert-driven weighting schemes and often prioritize intrinsic site attributes, the IGPI operationalizes geosite value as an emergent spatial performance shaped by accessibility, perceptual exposure, and digital mediation. By integrating these dimensions, the index addresses a key limitation in existing frameworks such as IELIG and GtRAM, which rarely incorporate spatial performance metrics that directly influence visitor behavior and operational feasibility (Carrión-Mero et al. 2021; Keskin Citioglu and Arca 2023). The coupling of IGPI rankings with network-based route optimization further advances beyond

static suitability mapping. Using OpenStreetMap-derived networks processed through OSMnx, travel-time matrices were constructed and optimized using Dijkstra's shortest-path algorithm and a TSP formulation, with walking effort modeled via Tobler's hiking function (Dijkstra 1959; Tobler 1993; Boeing 2017). The resulting full and reduced vehicular circuits—requiring 465 and 308 min, respectively—demonstrate that analytically derived georoutes can be translated into feasible day-scale itineraries while maintaining geomorphological coherence. This integration closes a common gap in geotourism studies between site prioritization and operational route design (Prameshwori et al. 2021; Gavilanes Montoya et al. 2021).

Methodological Robustness and Transferability

From a methodological standpoint, the study demonstrates the applicability of open-source GIS and network modelling for rural geoheritage assessment. Nonetheless, OpenStreetMap may include positional inaccuracies and incomplete seasonal/minor roads; Google-based digital visibility captures user activity rather than verified visitation, which can over-represent urban-proximate sites. Similar data issues have been reported in regional accessibility work (Gungor et al. 2025). Tourism-focused network analyses offer routing and service-area functions but are typically single-mode and static (Prameshwori et al. 2021); suitability-mapping frameworks effectively rank potential but often stop short of multi-modal, time-cost optimization (Louz et al. 2025). Future research should therefore integrate seasonal road accessibility, high-resolution DEM/LIDAR for micro-topography and viewshed obstruction, and participatory mapping for local validation. Among these, seasonal accessibility refinements are expected to have the greatest impact in high-altitude sectors such as Göze and Posof. The national-scale AHP findings (Bilgiç et al. 2024) also argue for reporting weight-sensitivity to increase transparency and transferability.

Policy and Governance Implications

Beyond methods, the Ardahan case carries broader implications for regional development and geopark governance. Spatial connectivity and digital representation emerge as mutually reinforcing mechanisms capable of transforming isolated geosites into a cohesive territorial system that fosters local entrepreneurship, diversifies rural incomes, and stabilizes population trends. Experiences from the Altar Volcano show that pairing theme-based routes with facilities/activities can channel visitor flows and raise site visibility (Gavilanes Montoya et al. 2021). Similar synergies were observed in the Seferihisar Cittaslow initiative (Yalçinkaya

et al. 2025), where GIS-informed routing supported slow-city principles. Web-based heritage information systems can sustain long-term geoheritage management by institutionalizing data updates and participation (Arca et al. 2018). In Ardahan, combining corridor-driven GIS design with digital outreach would reinforce both the practical and symbolic pillars of geopark candidacy, aligning with Türkiye's National Sustainable Tourism Strategy (Ministry of Culture and Tourism 2022).

Limitations

This study has several limitations that should be considered when interpreting the results. First, seasonal factors such as snow cover, temporary road closures, and grazing-related access constraints were not explicitly modelled. Therefore, the IGPI values and optimized georoutes primarily reflect summer-season accessibility conditions. Second, although accessibility classes were refined through field observations, the framework does not incorporate participatory inputs from local stakeholders, which represents a socio-institutional limitation given the governance principles of UNESCO Global Geoparks. Third, route optimization was based on a deterministic TSP formulation, which does not capture stochastic or real-time uncertainties related to weather, infrastructure disruptions, or visitor preferences. In addition, digital visibility indicators derived from Google Maps are influenced by platform-specific algorithms that may privilege urban or highly frequented locations, meaning that digitally invisible geosites may reflect representational bias rather than limited geoheritage value. Finally, while equal weighting in the IGPI was intentionally adopted to ensure methodological neutrality, it remains a simplifying assumption, and future studies could explore alternative weighting schemes to reflect context-specific or stakeholder-driven priorities.

Synthesis and Outlook

Integrating accessibility, visibility, and digital engagement yields a replicable framework for geosite prioritization and georoute planning in rural regions. The synthesis bridges the gap between geoheritage inventory and operational geotourism management, producing insights applicable to policy, education, and planning. Identified accessibility corridors, high-visibility nodes, and participation gradients collectively outline a realistic roadmap for Ardahan's transition from a remote mountainous province to a spatially connected, digitally visible, and economically resilient geopark landscape. Echoing prior work on visibility-informed itineraries and tourism routing (Martínez-Graña et al. 2017; Herrera-Franco et al. 2022; Gavilanes Montoya et al. 2021)

and on multi-criteria suitability roadmaps (Louz et al. 2025) as well as national-scale accessibility evidence (Bilgiç et al. 2024), the results support an approach where data-driven decision-making converges with community-based sustainability.

Conclusion

This study introduces an integrated GIS-based framework for assessing the accessibility of potential geosites and designing optimal georoutes, demonstrated through the case of Ardahan Province, Türkiye. The results reveal a distinct spatial hierarchy: a cluster of highly accessible and visually dominant geosites forming a development axis along the Ardahan–Çıldır corridor, contrasted with remote mountain sites constrained by limited physical and digital visibility. This contrast underscores the need for balanced geotourism strategies that combine efficiency, inclusiveness, and conservation. By integrating physical accessibility, scenic value, and digital presence into the IGPI, the study bridges the gap between geoheritage assessment and operational route design. The equal weighting of IGPI components follows a neutrality principle designed to avoid subjective prioritization and to ensure transparent, reproducible decision-making—an approach particularly relevant for emerging geopark regions where planning priorities are still evolving. The identified high-priority sites—Ardahan Castle, Şeytan Castle, Çıldır and Aktaş Lakes, Akçakale Island, and the Colchic Sector Forests—represent the most feasible core for geopark zoning and visitor management. Optimized route modelling further demonstrates how data-driven planning can enhance spatial connectivity, reduce environmental impact, and strengthen visitors' experience. Beyond its regional relevance, the proposed framework contributes to broader geotourism and geopark policy by offering a replicable decision-support tool grounded in open-source and participatory data. Strengthening the synergy between spatial accessibility and digital visibility can foster sustainable rural revitalization, promote community engagement, and reinforce the visibility of underrepresented geoheritage landscapes. This aligns with national and international objectives for rural development, showing that targeted infrastructure investment and digital interpretation can function as complementary drivers of territorial cohesion. In operational terms, the findings point to several immediate steps that regional authorities can take to translate analytical insights into practice. First, improving signage, wayfinding, and small-scale visitor infrastructure along the Çıldır–Kura corridor would consolidate its role as Ardahan's primary geotourism spine and enhance visitor flow management. Second, developing thematic

trekking products in geomorphologically distinctive yet low-accessibility sectors—particularly in Posof, Göze, and the Sahara uplands—would diversify visitor experiences while easing concentration pressures on the main corridor. Third, expanding the digital presence of geosites through coordinated online mapping initiatives, QR-based interpretive tools, and community-supported content creation could rapidly increase the visibility and discoverability of currently underrepresented sites. Taken together, these steps would operationalize the study’s framework and support a phased, corridor-based approach to sustainable geopark development in Ardahan. Future research should focus on integrating seasonal mobility data, field validation, and community-based mapping to support adaptive and inclusive geopark governance.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12371-026-01270-1>.

Author Contributions SD: Conceptualization, methodology design, data curation, GIS analysis, visualization, and writing—original draft preparation, review, and editing. MD: Supervision, validation, and critical revision of the manuscript for important intellectual content. Both authors have read and approved the final version of the manuscript.

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declarations

Conflict of interest The authors declare no competing interests.

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